

Lidocaine (systemic): Drug information - UpToDate

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For additional information see "Lidocaine (local and regional anesthetic) and (systemic): Patient drug information" and "Lidocaine (systemic): Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Lidomark 1/5 [DSC];
- Lidomark 2/5 [DSC];
- ReadySharp Lidocaine [DSC];
- Xylocaine;
- Xylocaine MPF +RFID;
- Xylocaine-MPF;
- Xylocaine-MPF +RFID

Brand Names: Canada

- Xylocaine;
- Xylocaine Plain;
- Xylocard

Pharmacologic Category

- Antiarrhythmic Agent, Class Ib;
- Local Anesthetic

Dosing: Adult

Interstitial cystitis

Interstitial cystitis (bladder pain syndrome) (off-label use):

Note: Various dosage regimens of alkalized lidocaine alone or with heparin (20,000 to 50,000 units) have been used. When lidocaine and heparin are mixed, there is a risk of precipitation if proper alkalization does not occur. Prior to administration, lidocaine stability and pH should be determined after the components have been mixed (Ref).

Intravesical instillation: Single instillation: Single bladder installation of lidocaine (200 mg)/heparin (50,000 units)/sodium bicarbonate (420 mg) in 15 mL of sterile water, instilled into the bladder via catheter and allowed to dwell for 30 minutes before drainage (Ref).

Intravesical instillation: Weekly instillation: Weekly bladder instillations for 12 consecutive weeks with lidocaine 4% (5 mL [200 mg])/heparin (20,000 units)/sodium bicarbonate 7% (25 mL [1,750 mg]), instilled into an empty bladder via catheter and allowed to dwell for 30 minutes before drainage (Ref).

Intravesical instillation: Daily instillation: Daily bladder instillations for 5 days with liaocaine (200 mg)/sodium bicarbonate 8.4% solution (final volume of 10 mL), instilled into an empty bladder and allowed to dwell for 1 hour before drainage (Ref).

Intraosseous, infusion pain

Intraosseous, infusion pain (off-label use): Note: May consider in a conscious patient to decrease pain prior to the infusion of fluid, medications, or blood products through the intraosseous cannula (Ref).

Liaocaine 2% preservative-free: **Intraosseous:** Initial dose: 20 or 40 mg over 1 to 2 minutes; may administer 1 or 2 additional doses of 20 mg; follow each dose with an NS flush; maximum total dose not established (Ref).

Local or regional anesthesia

Local or regional anesthesia: Note: Dosage varies with procedure, degree of anesthesia needed, vascularity of tissue, duration of anesthesia required, and physical condition of patient. Calculate weight-based doses using ideal body weight (Ref). General maximum single dose (unless noted otherwise below): 4.5 mg/kg/dose not to exceed 250 to 300 mg; reduced dosages are recommended for patients at high risk for local anesthetic systemic toxicity (Ref). Refer to institutional protocols; doses presented below are examples of recommended volumes and concentrations. Use the lowest concentration and smallest dose to achieve desired result; avoid rapid administration; inject slowly in fractional doses.

Caudal:

Note: The maximum recommended dose should not be administered at intervals of <90 minutes. A 3 mL test dose of liaocaine 1.5% (with epinephrine 1:200,000) given prior to injecting the total volume of liaocaine is recommended for a caudal epidural block. Use of a test dose may serve as a warning of unintentional intravascular or intrathecal injection.

Obstetrical **analgesia** (preservative free): Liaocaine 1%: 20 to 30 **mL**; total dose: 200 to 300 mg.

Surgical **anesthesia** (preservative free): Liaocaine 1.5%: 15 to 20 **mL**; total dose: 225 to 300 mg.

Epidural:

Note: Dosage varies with the number of dermatomes to be anesthetized (generally 2 to 3 mL of the indicated concentration per dermatome). The maximum recommended dose should not be administered at intervals of <90 minutes. A 3 mL test dose of liaocaine 1.5% (with epinephrine 1:200,000) given prior to injecting the total volume of liaocaine is recommended for a lumbar epidural block. Use of a test dose may serve as a warning of unintentional intravascular or intrathecal injection.

Thoracic epidural anesthesia (preservative free): Liaocaine 1%: 20 to 30 **mL**; total dose: 200 to 300 mg.

Lumbar epidural **anesthesia** (preservative free):

Liaocaine 1.5%: 15 to 20 **mL**; total dose: 225 to 300 mg.

Liaocaine 2%: 10 to 15 **mL**; total dose: 200 to 300 mg.

Lumbar epidural **analgesia** (preservative free): Liaocaine 1%: 25 to 30 **mL**; total dose: 250 to 300 mg.

Infiltration (local anesthesia):

Cutaneous injection (eg, SUBQ or intradermal): Liaocaine 0.5% or 1%: 1 to 60 **mL**; total dose: 5 to 300 mg.

IV regional anesthesia (eg, Bier block):

Note: Only the single-dose vial containing liaocaine hydrochloride 0.5% injection (without epinephrine) should be used for IV regional anesthesia (Ref).

IV (after tourniquet occlusion of limb to prevent systemic exposure): Lidocaine 0.5%: 10 to 60 **mL**; total dose: 50 to 300 mg. Usual dosage range: 30 to 50 **mL** (150 to 250 mg) for upper extremity blocks; maximum dose: 4 mg/kg/dose (Ref).

Nerve block, peripheral (regional anesthesia):

Brachial: Lidocaine 1.5%: 15 to 20 **mL**; total dose: 225 to 300 mg.

Dental: Lidocaine 2%: 1 to 5 **mL**; total dose: 20 to 100 mg.

Intercostal: Lidocaine 1%: 3 **mL**; total dose: 30 mg.

Paravertebral: Lidocaine 1%: 3 to 5 **mL**; total dose: 30 to 50 mg.

Pudendal (each side): Lidocaine 1%: 10 **mL**; total dose: 100 mg.

Nerve block, sympathetic (regional anesthesia):

Cervical: Lidocaine 1%: 5 **mL**; total dose: 50 mg.

Lumbar: Lidocaine 1%: 5 to 10 **mL**; total dose: 50 to 100 mg.

Paracervical: Obstetrical analgesia (each side): Lidocaine 1%: 10 **mL**; total dose: 100 mg, wait 5 minutes before administering on other side; maximum dose: 200 mg (100 mg on each side) in a 90 minute period.

Retrobulbar: Lidocaine 4%: 3 to 5 **mL**; total dose: 120 to 200 mg; a portion of this dose is injected retrobulbarly and the remainder may be used to block the facial nerve.

Perioperative pain

Perioperative pain (alternative agent) (off-label use):

Note: For use as an alternative to regional analgesia/anesthesia strategies (eg, epidural, nerve blocks) in patients who undergo major surgery (eg, thoracic, lumbar, open abdominal surgery) (Ref). Not recommended for use in patients with actual body weight <40 kg (Ref). The optimal dosing regimen has yet to be defined. Refer to institutional policies and procedures.

IV: 1 to 1.5 mg/kg (using ideal body weight [IBW]) given as a bolus near the time of induction of anesthesia followed by an infusion of 1.5 to 2 mg/kg/**hour** (using IBW; maximum: 120 mg/**hour**). In general, discontinue infusion at the time of skin closure; however, if patient is expected to have difficulty with pain control (eg, opioid tolerant) or opioid avoidance is prudent (eg, obstructive sleep apnea), may continue the infusion postoperatively for up to 24 hours (Ref). Rarely, postoperative infusions may be continued beyond 24 hours at 50% of the previous infusion rate if in concert with an appropriate specialist consultant and within the scope of relevant hospital guidelines (Ref).

Ventricular arrhythmias

Ventricular arrhythmias:

Sudden cardiac arrest due to ventricular fibrillation or pulseless ventricular tachycardia (unresponsive to CPR, defibrillation, epinephrine) (off-label use):

Bolus dose:

IV/intraosseous: 1 to 1.5 mg/kg; repeat with 0.5 to 0.75 mg/kg every 5 to 10 minutes as necessary; a typical dose is 50 to 100 mg; maximum cumulative dose: 3 mg/kg or up to 300 mg within 1 hour (Ref).

Endotracheal (off-label route; **only use when IV or intraosseous access cannot be established**): 2 to 3.75 mg/kg; before administration dilute in 5 to 10 mL NS or sterile water. **Note:** Absorption is greater with sterile water and results in less impairment of PaO₂ (Ref).

Continuous infusion:

Note: Even if lidocaine was not administered during resuscitation but return of spontaneous circulation was achieved, may still consider a lidocaine infusion to prevent recurrent arrhythmia (Ref).

IV/intraosseous : 1 to 4 mg/minute; if used beyond 24 hours, the rate of elimination can be prolonged and dose reduction may be necessary; monitor serum concentrations and signs or symptoms of toxicity to guide dose adjustment; maximum cumulative dose: 300 mg within 1 hour (Ref).

Ventricular tachycardia, hemodynamically stable, sustained monomorphic:

Bolus dose:

IV: 1 to 1.5 mg/kg; repeat with 0.5 to 0.75 mg/kg every 5 to 10 minutes as necessary; a typical dose is 50 to 100 mg; maximum cumulative dose: 3 mg/kg or up to 300 mg within 1 hour (Ref).

Continuous infusion:

Note: Consider use if arrhythmia recurs (ie, becomes incessant) after bolus dosing (Ref).

IV: 1 to 4 mg/minute; if used beyond 24 hours, the rate of elimination can be prolonged and dose reduction may be necessary; monitor serum concentrations and signs or symptoms of toxicity to guide dose adjustment; maximum cumulative dose: 300 mg within 1 hour (Ref).

Dosing: Kidney Impairment: Adult

eGFR <30 mL/minute/1.73 m²: Administer lower maintenance infusion rate with close monitoring for toxicity.

Dosing: Liver Impairment: Adult

Administer lower maintenance infusion rate with close monitoring for toxicity.

Dosing: Older Adult

Refer to adult dosing.

Perioperative pain (alternative agent) (off-label use):

Note: For use in patients who undergo thoracic or lumbar surgery or open abdominal surgery when continuous regional analgesia (eg, epidural, continuous peripheral nerve block) is not used due to contraindication, refusal, or failure (Ref). Not recommended for use in patients with actual body weight <40 kg (Ref). The optimal dosing regimen has yet to be defined. Refer to institutional policies and procedures.

IV: 1 to 1.5 mg/kg (using ideal body weight [IBW]) given as a bolus near the time of induction of anesthesia followed by an infusion of 1 mg/kg/hour (using IBW). In general, discontinue infusion at the time of skin closure; however, if patient is expected to have difficulty with pain control (eg, opioid tolerant) or opioid avoidance is prudent (eg, obstructive sleep apnea), may continue the infusion postoperatively for up to 24 hours (Ref). Rarely, postoperative infusions may be continued beyond 24 hours at 50% of the previous infusion rate if in concert with an appropriate specialist consultant and within the scope of relevant hospital guidelines (Ref).

Dosing: Pediatric

(For additional information see "Lidocaine (systemic): Pediatric drug information")

Ventricular fibrillation or pulseless ventricular tachycardia, shock refractory

Ventricular fibrillation (VF) or pulseless ventricular tachycardia (VT), shock refractory:

Infants, Children, and Adolescents:

Loading dose:

IV, Intraosseous: 1 mg/kg/dose; follow with continuous IV infusion; may administer second bolus if delay between initial bolus and start of infusion is >15 minutes (Ref).

Endotracheal: Loading dose: 2 to 3 mg/kg/dose; flush with ≥5 mL of NS and follow with 5 assisted manual ventilations (Ref).

Continuous IV infusion: 20 to 50 mcg/kg/minute (Ref). **Note:** Per manufacturer, do not exceed 20 mcg/kg/minute in patients with shock, hepatic disease, cardiac arrest, or congestive heart failure.

Anesthesia, local injectable

Anesthesia, local injectable: Dose varies with procedure, degree of anesthesia needed, vascularity of tissue, duration of anesthesia required, and physical condition of patient.

Cutaneous infiltration: Children and Adolescents: Typically solutions with concentration <2% should be used (allow for larger volumes); maximum dose: 5 mg/kg/dose not to exceed the recommended adult maximum dose of 300 mg/dose; do not repeat within 2 hours (Ref).

Intraosseous line or infusion pain: Infants, Children, and Adolescents: Lidocaine 1% or 2% preservative-free solution: Intraosseous: Initial dose: 0.5 mg/kg over 1 to 2 minutes, maximum dose: 40 mg/dose; follow with NS flush; immediately following the NS flush, if needed, may administer a second dose (50% of initial dose) over 30 to 60 seconds (usual adult maximum repeat dose: 20 mg/dose); if discomfort reoccurs, may repeat doses at a maximum frequency of every 45 minutes during intraosseous access; maximum total dose not established, some centers suggest that dose should not exceed 3 mg/kg per 24 hours (Ref).

Dosing: Kidney Impairment: Pediatric

Infants, Children, and Adolescents:

Note: Accumulation of metabolites may be increased in kidney impairment.

eGFR <30 mL/minute/1.73 m²: IV: Administer maintenance infusion at a lower rate; monitor closely for toxicity.

Dosing: Liver Impairment: Pediatric

Infants, Children, and Adolescents: Use with caution; consider dose reduction. Monitor lidocaine concentrations closely and adjust infusion rate as necessary; consider alternative therapy.

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified. Adverse effects may result from high serum concentrations due to rapid absorption, unintentional intravascular administration, unapproved use, or overdose.

Postmarketing:

Cardiovascular: Bradycardia, circulatory shock, hypotension

Gastrointestinal: Vomiting

Hypersensitivity: Hypersensitivity reaction (including nonimmune anaphylaxis)

Nervous system: Abnormal sensory symptoms (including sensation of heat, sensation of cold), apprehension, confusion, dizziness, drowsiness, euphoria, loss of consciousness, nervousness, numbness, seizure, tremor, twitching

Ophthalmic: Blurred vision, diplopia

Otic: Tinnitus

Respiratory: Respiratory depression

Contraindications

Hypersensitivity to lidocaine or any component of the formulation; hypersensitivity to another local anesthetic of the amide type; Adam-Stokes syndrome; Wolff-Parkinson-White syndrome; severe degrees of SA, AV, or intraventricular heart block (except in patients with a functioning artificial pacemaker); premixed injection may contain corn-derived dextrose and its use is contraindicated in patients with allergy to corn or corn-related products.

Canadian labeling: Additional contraindications (not in US labeling): Hypersensitivity to ester local anesthetics (paraben-containing solutions only); supraventricular arrhythmias; severe myocardial depression; antimicrobial preservative-containing solutions should not be used intra- or retro-ocularly

or for epidural or spinal anesthesia or any route that would introduce solution into the cerebrospinal fluid or in doses ≥ 15 mL for other types of blockades.

Warnings/Precautions

Concerns related to adverse effects:

- Intra-articular infusion related chondrolysis: Continuous intra-articular infusion of local anesthetics after arthroscopic or other surgical procedures is **not** an approved use; chondrolysis (primarily in the shoulder joint) has occurred following infusion, with some cases requiring arthroplasty or shoulder replacement.
- Methemoglobinemia: Has been reported with local anesthetics; clinically significant methemoglobinemia requires immediate treatment along with discontinuation of the anesthetic and other oxidizing agents. Onset may be immediate or delayed (hours) after anesthetic exposure. Patients with glucose-6-phosphate dehydrogenase deficiency, congenital or idiopathic methemoglobinemia, cardiac or pulmonary compromise, exposure to oxidizing agents or their metabolites, or infants <6 months are more susceptible and should be closely monitored for signs and symptoms of methemoglobinemia (eg, cyanosis, headache, rapid pulse, shortness of breath, lightheadedness, fatigue).

Disease-related concerns:

- Hepatic dysfunction: Use extreme caution in patients with severe hepatic dysfunction; may have increased risk of lidocaine toxicity.

Dosage form specific issues:

- Benzyl alcohol and derivatives: Some dosage forms may contain benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome") in neonates; the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP ["Inactive" 1997]; CDC 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol with caution in neonates. See manufacturer's labeling.
- Injectable anesthetic: Follow appropriate administration techniques so as not to administer any intravascularly. Solutions containing antimicrobial preservatives should not be used for epidural or spinal anesthesia. Some solutions contain a bisulfite; avoid in patients who are allergic to bisulfite. Resuscitative equipment, medicine and oxygen should be available in case of emergency. Adjust the dose for elderly, pediatric, acutely ill, and debilitated patients.
- Intravenous: Constant ECG monitoring is necessary during IV administration. Use cautiously in hepatic impairment, HF, marked hypoxia, severe respiratory depression, hypovolemia, history of malignant hyperthermia, or shock. Increased ventricular rate may be seen when administered to a patient with atrial fibrillation. Use is contraindicated in patients with Wolff-Parkinson-White syndrome and severe degrees of SA, AV, or intraventricular heart block (except in patients with a functioning artificial pacemaker). Correct electrolyte disturbances, especially hypokalemia or hypomagnesemia, prior to use and throughout therapy. Correct any underlying causes of ventricular arrhythmias. Monitor closely for signs and symptoms of CNS toxicity. The elderly may be prone to increased CNS and cardiovascular side effects. Reduce dose in hepatic dysfunction and CHF.

Other warnings/precautions:

- CAST trial: In the Cardiac Arrhythmia Suppression Trial (CAST), recent (>6 days but <2 years ago) myocardial infarction patients with asymptomatic, non-life-threatening ventricular arrhythmias did not benefit and may have been harmed by attempts to suppress the arrhythmia with flecainide or encainide. An increased mortality or nonfatal cardiac arrest rate (7.7%) was seen in the active treatment group compared with patients in the placebo group (3%). The applicability of the CAST results to other populations is unknown. Antiarrhythmic agents should be reserved for patients with life-threatening ventricular arrhythmias.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Kit, Injection, as hydrochloride:

Lidomark 1/5: 1% [DSC]

Lidomark 2/5: 2% [DSC]

ReadySharp Lidocaine: 1% [DSC]

Solution, Injection, as hydrochloride:

Xylocaine: 0.5% (50 mL); 1% (20 mL, 50 mL); 2% (10 mL, 20 mL, 50 mL) [contains methylparaben (methyl p-hydroxybenzoate)]

Xylocaine MPF +RFID: 1% (2 mL) [methylparaben free]

Xylocaine-MPF +RFID: 2% (2 mL, 5 mL) [methylparaben free]

Generic: 0.5% (50 mL); 1% (2 mL, 5 mL, 10 mL, 20 mL, 50 mL); 2% (2 mL, 5 mL, 10 mL, 20 mL, 50 mL)

Solution, Injection, as hydrochloride [preservative free]:

Xylocaine-MPF: 0.5% (50 mL); 1% (2 mL, 5 mL, 10 mL, 30 mL); 1.5% (10 mL, 20 mL); 2% (2 mL, 5 mL, 10 mL) [methylparaben free]

Generic: 0.5% (50 mL); 1% (2 mL, 5 mL, 10 mL, 30 mL); 1.5% (20 mL); 2% (2 mL, 5 mL, 10 mL, 20 mL); 4% (5 mL)

Solution, Intravenous, as hydrochloride:

Generic: 0.4% [4 mg/mL] (250 mL, 500 mL); 0.8% [8 mg/mL] (250 mL)

Solution, Intravenous, as hydrochloride [preservative free]:

Generic: 100 mg/5 mL (5 mL)

Solution Prefilled Syringe, Injection, as hydrochloride [preservative free]:

Generic: 2% (5 mL)

Solution Prefilled Syringe, Intravenous, as hydrochloride:

Generic: 50 mg/5 mL (5 mL)

Solution Prefilled Syringe, Intravenous, as hydrochloride [preservative free]:

Generic: 50 mg/5 mL (5 mL [DSC]); 100 mg/5 mL (5 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Lidocaine HCl (Cardiac) PF Intravenous)

100 mg/5 mL (per mL): \$0.90

Solution (Lidocaine HCl (PF) Injection)

0.5% (per mL): \$0.10

1% (per mL): \$0.24 - \$19.60

1.5% (per mL): \$0.61

2% (per mL): \$0.27 - \$1.11

4% (per mL): \$1.16 - \$2.13

Solution (Lidocaine HCl Injection)

0.5% (per mL): \$0.11

1% (per mL): \$0.10 - \$0.37

2% (per mL): \$0.14 - \$0.47

Solution (Lidocaine in D5W Intravenous)

4 mg/mL 5% (per mL): \$0.02

8 mg/mL 5% (per mL): \$0.04

Solution (Xylocaine Injection)

0.5% (per mL): \$0.13

1% (per mL): \$0.10

2% (per mL): \$0.10

Solution (Xylocaine MPF +RFID Injection)

1% (per mL): \$1.60

Solution (Xylocaine-MPF +RFID Injection)

2% (per mL): \$1.94

Solution (Xylocaine-MPF Injection)

0.5% (per mL): \$0.35

1% (per mL): \$0.38

1.5% (per mL): \$0.92

2% (per mL): \$0.34

Solution Prefilled Syringe (Lidocaine HCl (Cardiac) PF Intravenous)

50 mg/5 mL (per mL): \$3.60

100 mg/5 mL (per mL): \$1.74

Solution Prefilled Syringe (Lidocaine HCl Injection)

100 mg/5 mL (per mL): \$2.36

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection, as hydrochloride:

Xylocaine: 0.5% (20 mL)

Xylocaine: 1% (20 mL, 50 mL); 2% (20 mL, 50 mL) [contains methylparaben (methyl p-hydroxybenzoate)]

Xylocaine Plain: 1% (2 mL, 5 mL, 10 mL); 2% (2 mL, 5 mL, 10 mL)

Generic: 10 mg/mL (5 mL, 10 mL, 20 mL, 50 mL); 1% (2 mL, 5 mL, 10 mL, 20 mL, 50 mL); 2% (2 mL, 5 mL, 10 mL, 20 mL, 50 mL)

Solution, Intravenous, as hydrochloride:

Xylocard: 2% [20 mg/mL] (5 mL)

Generic: 100 mg/5 mL (5 mL); 0.4% [4 mg/mL] (250 mL, 500 mL); 2% [20 mg/mL] (5 mL)

Solution Prefilled Syringe, Intravenous, as hydrochloride:

Generic: 100 mg/5 mL (5 mL)

Administration: Adult

IV (systemic):

Bolus: According to the manufacturer, may administer at 25 to 50 mg/minute. In the setting of cardiac arrest (eg, ventricular fibrillation or pulseless ventricular tachycardia), may be infused rapidly into a peripheral vein (Ref).

Continuous infusion: After initial bolus dosing, may administer as a continuous infusion; refer to indication-specific infusion rates in dosing for detailed recommendations. In the setting of cardiac arrest, infusion may be initiated once patient has return of spontaneous circulation resulting from lidocaine administration. Local thrombophlebitis may occur in patients receiving prolonged IV infusions.

IV (regional anesthesia): Prior to injection, verify proper tourniquet occlusion of limb to prevent systemic exposure. Administer slowly over ≥ 90 seconds (Ref).

Cutaneous infiltration (local anesthesia) or perineural infiltration (regional anesthesia): Administer as single or multiple injections, under ultrasound guidance whenever possible; may administer incremental injections or continuously through an indwelling catheter. Avoid rapid injection. Avoid intravascular injections; aspirate syringe prior to each injection; the needle must be repositioned until no return of blood can be elicited by aspiration; however, absence of blood in the syringe does not guarantee that intravascular injection has been avoided (Ref).

The On-Q infusion pump is used to slowly administer local anesthetics (eg, bupivacaine, lidocaine, ropivacaine) to or around surgical wound sites and/or in close proximity to peripheral nerves for postoperative analgesia. When infused directly into the shoulder, destruction of articular cartilage (chondrolysis) has occurred. On-Q pumps should never be placed directly into any joint (Ref).

Endotracheal (off-label administration route): Dilute in NS or sterile water. Absorption is greater with sterile water and results in less impairment of PaO₂ (Ref). Stop compressions, spray drug quickly down tube. Flush with 5 mL of NS and follow immediately with several quick insufflations and continue chest compressions.

Epidural and spinal anesthesia: Do not use solutions containing preservatives. Do not puncture areas of the skin with signs of infection. Intravascular injections should be avoided; aspiration should be performed prior to administration; the needle must be repositioned until no return of blood can be elicited by aspiration; however, absence of blood in the syringe does not guarantee that intravascular injection has been avoided.

Intraosseous (IO; off-label administration route): Intraosseous administration is a reasonable alternative when quick IV access is not feasible (Ref).

Intravesical instillation (off-label use): Various regimens of alkalized lidocaine (with or without heparin) have been instilled into the bladder (Ref).

Preparation for Administration: Adult

Local infiltration: Buffered lidocaine for injectable local anesthetic may be prepared to decrease pain on injection: Add 1 mL of sodium bicarbonate 8.4% to 9 mL of lidocaine 1% (Hockett 2021) **or** add 1 mL of sodium bicarbonate 8.4% to 10 mL lidocaine 1% or 2% (Pascuet 2009). Alternatively, add 2 mL of sodium bicarbonate 8.4% to 18 mL of lidocaine 1% (Ref).

IV (systemic):

Bolus: The injectable solution of 10 or 20 mg/mL may be administered undiluted.

Continuous IV infusion: Dilute in D5W or other compatible solution to a concentration of 1,000 to 8,000 mcg/mL (1 to 8 **mg**/mL) (Ref); premix solution available for 4,000 and 8,000 mcg/mL (4 and 8 **mg**/mL).

Usual Infusion Concentrations: Adult

Note: Premixed solutions available.

IV infusion: 1,000 mg in 250 mL (concentration: 4 mg/mL) **or** 2,000 mg in 250 mL (concentration: 8 mg/mL) of D₅W.

Administration: Pediatric

Endotracheal: May administer dose undiluted **or** may further dilute in 1 to 5 mL NS based on patient size; follow with a flush of ≥5 mL of NS after endotracheal administration; follow with 5 assisted manual ventilations (Ref).

Parenteral:

IV push, Intraosseous:

Ventricular fibrillation or pulseless ventricular tachycardia: The manufacturer recommends that the rate of administration should not exceed 0.7 mg/kg/minute or 50 mg/minute, whichever is less; however, during acute situations, administration by rapid IV push has been used by some clinicians in practice.

Neonatal seizures: Loading doses are usually administered over 10 minutes (Ref).

Continuous IV infusion: Administer via an infusion pump.

SUBQ/local infiltration: May use plain or buffered lidocaine; for circumcisions, use a 23- to 27-gauge needle for administration (Ref).

Preparation for Administration: Pediatric

Endotracheal: May dilute in 1 to 5 mL NS based on patient size (Ref).

Parenteral:

IV: Continuous IV infusion: Dilute in D5W or other compatible solution to a final concentration of 1 to 8 mg/mL (Ref); premixed IV solutions are available (4 mg/mL, 8 mg/mL). For pediatric patients, ASHP recommends standard concentrations of 4 mg/mL and 8 mg/mL (for cardiac indications only) (Ref).

Local infiltration: Buffered lidocaine for injectable local anesthetic may be prepared: Add 2 mL of sodium bicarbonate 8.4% to 18 or 20 mL of lidocaine 1% (Ref).

Usual Infusion Concentrations: Pediatric

Note: Premixed solutions available.

IV infusion: 4 mg/mL, 8 mg/mL.

Storage/Stability

Injection: Store at controlled room temperature. For products with an overwrap, do not remove overwrap until ready for use.

Use: Labeled Indications

Local or regional anesthesia: Local and regional anesthesia by infiltration, nerve block, epidural, or spinal techniques.

Ventricular tachycardia, hemodynamically stable, sustained monomorphic: Acute treatment of ventricular arrhythmias (eg, due to acute myocardial infarction [MI] or during cardiac manipulation [eg, cardiac surgery]).

Use: Off-Label: Adult

Interstitial cystitis (bladder pain syndrome); Intraosseous, infusion pain; Perioperative pain; Sudden cardiac arrest due to ventricular fibrillation or pulseless ventricular tachycardia (unresponsive to CPR, defibrillation, and epinephrine)

Medication Safety Issues

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (antiarrhythmic agent, IV; epidural and intrathecal medications) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

International issues:

Lidosen [Italy] may be confused with Lincocin brand name for lincomycin [US, Canada, and multiple international markets]; Lodosyn brand name for carbidopa [US].

Metabolism/Transport Effects

Substrate of CYP1A2 (Major), CYP2A6 (Minor), CYP2B6 (Minor), CYP2C9 (Minor), CYP3A4 (Major with inhibitors), CYP3A4 (Minor with inducers); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Amiodarone: May increase serum concentrations of Lidocaine (Systemic). *Risk C: Monitor*

Articaine: May increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

Broccoli: May decrease serum concentrations of CYP1A2 Substrates (High risk with Inducers). *Risk C: Monitor*

BUPIvacaine (Liposomal): Lidocaine (Systemic) may increase adverse/toxic effects of BUPIvacaine (Liposomal). Management: Liposomal bupivacaine should not be administered with lidocaine. Liposomal bupivacaine may be administered 20 minutes or more after the administration of lidocaine. Lidocaine may be administered 96 hours or more after liposomal bupivacaine administration. *Risk D: Consider Therapy Modification*

BUPIvacaine: Local Anesthetics may increase adverse/toxic effects of BUPIvacaine. Management: Avoid using any additional local anesthetics within 96 hours after insertion of the bupivacaine implant (Xaracoll) or bupivacaine and meloxicam periarticular solution (Zynrelef) or within 168 hours after subacromial infiltration (Posimir brand). *Risk C: Monitor*

Cannabis: May decrease serum concentrations of CYP1A2 Substrates (High risk with Inducers). *Risk C: Monitor*

Cimetidine: May increase serum concentrations of Lidocaine (Systemic). *Risk C: Monitor*

Clofazimine: May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). *Risk C: Monitor*

CYP1A2 Inducers (Moderate): May decrease serum concentrations of Lidocaine (Systemic). *Risk C: Monitor*

CYP1A2 Inhibitors (Strong): May increase serum concentrations of Lidocaine (Systemic). *Risk C: Monitor*

CYP3A4 Inducers (Strong): May decrease serum concentrations of Lidocaine (Systemic). *Risk C: Monitor*

CYP3A4 Inhibitors (Moderate): May increase serum concentrations of Lidocaine (Systemic). CYP3A4 Inhibitors (Moderate) may increase active metabolite exposure of Lidocaine (Systemic). Specifically, concentrations of monoethylglycinexylidide (MEGX) may be increased. *Risk C: Monitor*

CYP3A4 Inhibitors (Strong): May increase serum concentrations of Lidocaine (Systemic). *Risk C: Monitor*

Dapsone (Topical): May increase adverse/toxic effects of Methemoglobinemia Associated Agents. *Risk C: Monitor*

Diazoxide Choline (Extended Release): May increase serum concentrations of CYP1A2 Substrates (High risk with Inhibitors). Management: Consider alternatives when possible, as product labeling states that coadministration of diazoxide choline and CYP1A2 substrates is not recommended. If combined, monitor for increased effects of the CYP1A2 substrate. *Risk D: Consider Therapy Modification*

Disopyramide: May increase arrhythmogenic effects of Lidocaine (Systemic). Disopyramide may increase serum concentrations of Lidocaine (Systemic). Specifically, the unbound/free fraction of lidocaine may be increased. *Risk C: Monitor*

Etravirine: May decrease serum concentrations of Lidocaine (Systemic). *Risk C: Monitor*

Fusidic Acid (Systemic): May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). Management: Consider avoiding this combination if possible. If required, monitor patients closely for increased adverse effects of the CYP3A4 substrate. *Risk D: Consider Therapy Modification*

Grapefruit Juice: May increase serum concentrations of Lidocaine (Systemic). Grapefruit Juice may increase active metabolite exposure of Lidocaine (Systemic). Specifically, concentrations of monoethylglycinexylidide (MEGX) may be increased. *Risk C: Monitor*

Hyaluronidase: May increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

Lacosamide: Lidocaine (Systemic) may increase adverse/toxic effects of Lacosamide. Specifically the risk for bradycardia, ventricular tachyarrhythmias, or a prolonged PR interval may be increased. *Risk C: Monitor*

Lidocaine (Topical): May increase adverse/toxic effects of Antiarrhythmic Agents (Class IB). *Risk C: Monitor*

Local Anesthetics: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Local Anesthetics. Specifically, the risk for methemoglobinemia may be increased. *Risk C: Monitor*

Methemoglobinemia Associated Agents: May increase adverse/toxic effects of Local Anesthetics. Specifically, the risk for methemoglobinemia may be increased. *Risk C: Monitor*

Mexiletine: May increase serum concentrations of Lidocaine (Systemic). Adverse/toxic effects of Lidocaine (Systemic) and Mexiletine may increase with coadministration. *Risk C: Monitor*

Neuromuscular-Blocking Agents: Local Anesthetics may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents. *Risk C: Monitor*

Nitric Oxide: May increase adverse/toxic effects of Methemoglobinemia Associated Agents. Combinations of these agents may increase the likelihood of significant methemoglobinemia. *Risk C: Monitor*

Phenobarbital-Primidone: May decrease serum concentrations of Lidocaine (Systemic). *Risk C: Monitor*

Prilocaine: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Prilocaine. Combinations of these agents may increase the likelihood of significant methemoglobinemia. Management: Monitor for signs of methemoglobinemia when prilocaine is used in combination with other agents associated with development of methemoglobinemia. Avoid use of these agents with prilocaine/lidocaine cream in infants less than 12 months of age. *Risk C: Monitor*

Primaquine: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Primaquine. Specifically, the risk for methemoglobinemia may be increased. Management: Avoid concomitant use of primaquine and other drugs that are associated with methemoglobinemia when possible. If combined, monitor methemoglobin levels closely. *Risk D: Consider Therapy Modification*

Propranolol: May increase serum concentrations of Lidocaine (Systemic). *Risk C: Monitor*

Pyridostigmine: May increase or decrease neuromuscular-blocking effects of Local Anesthetics. *Risk C: Monitor*

ROPivacaine: May increase adverse/toxic effects of Local Anesthetics. *Risk C: Monitor*

Sodium Nitrite: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Sodium Nitrite. Combinations of these agents may increase the likelihood of significant methemoglobinemia. *Risk C: Monitor*

Technetium Tc 99m Tilmanocept: Coadministration of Local Anesthetics and Technetium Tc 99m Tilmanocept may alter diagnostic results. Management: Avoid mixing and simultaneously co-injecting technetium Tc 99m tilmanocept with local anesthetics. This interaction does not appear to apply to other uses of these agents in combination. *Risk C: Monitor*

Tobacco (Smoked): May decrease serum concentrations of Lidocaine (Systemic). *Risk C: Monitor*

Pregnancy Considerations

Lidocaine and its metabolites cross the placenta and can be detected in the fetal circulation following maternal injection for anesthesia prior to delivery (Cavalli 2004; Mitani 1987).

Outcome data and decades of experience following maternal use of systemic lidocaine during pregnancy are available; a drug-associated risk of major birth defects, miscarriage, or adverse maternal or fetal outcomes has not been identified (manufacturer's labeling).

Lidocaine injection is approved for obstetric analgesia (eg, prior to epidural or spinal anesthesia). Intradermal administration is not detectable systemically; fetal exposure is not expected with maternal administration and is used to provide analgesia prior to episiotomy and during repair of obstetric lacerations (ACOG 209 2019; manufacturer's labeling). Lidocaine administered as perineal analgesia may result in greater absorption than administration by the epidural route (Cavalli 2004). Cumulative exposure from all routes of administration should be considered. The ACOG recommends that pregnant patients should not be denied medically necessary surgery regardless of trimester. If the procedure is elective, it should be delayed until after delivery (ACOG 775 2019).

Medications used for the treatment of cardiac arrest in pregnancy are the same as in the nonpregnant woman. Doses and indications should follow current Advanced Cardiovascular Life Support guidelines. Appropriate medications should not be withheld due to concerns of fetal teratogenicity (AHA [Jeejeebhoy 2015]).

Breastfeeding Considerations

Lidocaine is present in human milk.

- Data related to the presence of lidocaine in human milk are available following maternal administration of lidocaine via local regional anesthesia to 22 patients undergoing cesarean delivery. Patients received a mean dose of lidocaine 183 ± 104 mg. Lidocaine concentrations 2 hours after the injection were 0.86 ± 0.79 mcg/mL (human milk) and 0.94 ± 0.54 mcg/mL (maternal serum). Human

milk concentrations of lidocaine decreased over 12 hours. Adverse events were not observed in the breastfed infants (Ortega 1999).

- Lidocaine concentrations in human milk have been evaluated in lactating patients following use as a dental anesthetic (Giuliani 2001; Lebedevs 1993; Puente 2001). In one study, patients were administered lidocaine 2% (3.6 mL) (n=6), lidocaine 2% (4.5 mL) (n=1), or lidocaine 2% (7.2 mL) (n=1). Human milk was sampled 3 and 6 hours after the dose; maternal serum was sampled 2 hours after the dose. Lidocaine concentrations were 0.078 to 0.25 mcg/mL (human milk at 3 hours), 0.0283 to 0.107 mcg/mL (human milk at 6 hours), and 0.196 to 0.869 mcg/mL (maternal serum at 2 hours). The active metabolite monoethylglycinexylidide (MEGX) was also present in human milk and maternal serum at all sampling times. Human milk concentrations of MEGX were greatest 3 hours after the maternal dose (0.0333 to 0.143 mcg/mL). Infants were not breastfed in this study (Giuliani 2001).
- Lidocaine was detected in human milk following use in a tumescent liposuction procedure. The patient received a total of 4,200 mg throughout the procedure. Lidocaine concentrations were 0.55 mcg/mL (human milk) and 1.2 mcg/mL (maternal serum) at the expected peak plasma time, 18 hours after the surgery. The infant was not breastfed (Dryden 2000).
- Human milk was sampled in a patient following a lidocaine infusion for treatment of a maternal arrhythmia. The 86 kg lactating patient who had a history of mitral valve prolapse presented to the emergency room with heart palpitations and chest pain. A lidocaine 75 mg IV bolus was administered, followed by a second lidocaine 50 mg IV bolus after 5 minutes. A lidocaine 23 mcg/kg/minute infusion was started with the first bolus. Five hours after lidocaine was initiated, maternal serum concentrations were 2 mcg/mL; the infusion was then discontinued 2 hours later. Lidocaine human milk concentrations after the infusion was stopped were 0.8 mcg/mL (Zeisler 1986).
- Oral bioavailability of lidocaine to the breastfeeding infant is expected to be low (Lebedevs 1993; Ortega 1999).

Available guidelines consider lidocaine to be compatible with breastfeeding when used as an antiarrhythmic or local anesthetic (ABM [Reece-Stremtan 2017]; WHO 2002). Cumulative exposure from all routes of administration should be considered.

Dietary Considerations

Premixed injection may contain corn-derived dextrose and its use is contraindicated in patients with allergy to corn-related products.

Monitoring Parameters

LFTs, ECG; neurologic signs and symptoms of toxicity (eg, perioral tingling, tinnitus, light-headedness, restlessness); in patients requiring continuous infusions or therapy >24 hours, lidocaine serum concentrations if available and clinically relevant; consult individual institutional policies and procedures.

Reference Range

Note: These ranges act as a general guide only; the correlation between toxic symptoms and plasma lidocaine concentrations is not fully reliable. Vulnerable patients may experience toxicity at lower levels (Foo 2021; Samarin 2015).

Therapeutic: 1.5 to 5 mcg/mL (SI: 6.4 to 21.3 micromole/L).

Toxicity more common: >6 mcg/mL (SI: >25.6 micromole/L).

Mechanism of Action

Class Ib antiarrhythmic; suppresses automaticity of conduction tissue, by increasing electrical stimulation threshold of ventricle, His-Purkinje system, and spontaneous depolarization of the ventricles during diastole by a direct action on the tissues; blocks both the initiation and conduction of nerve impulses by decreasing the neuronal membrane's permeability to sodium ions, which results in inhibition of depolarization with resultant blockade of conduction

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Systemic administration: Single IV bolus dose: 45 to 90 seconds.

Duration: Systemic administration: Single IV bolus dose: 10 to 20 minutes.

Distribution: Systemic administration: Single IV bolus dose: 0.8 to 2.8 L/kg (Berkenstadt 1999); alterable by many patient factors; decreased in heart failure and liver disease; crosses blood-brain barrier.

Protein binding: Dependent on drug concentration; fraction bound decreases with increasing concentration. At 1 to 4 mcg/mL, 60% to 80% bound to α_1 acid glycoprotein.

Metabolism: 90% hepatic via CYP1A2 and CYP3A4; active metabolites monoethylglycinexylidide (MEGX) and glycinexylidide (GX) can accumulate and may cause CNS toxicity.

Half-life elimination: Systemic administration: Biphasic: Initial: ~8 minutes following single IV bolus (Burm 1988); Terminal: Infants, premature: 3.2 hours; Adults: 1.5 to 2 hours following single IV bolus (Burm 1988; manufacturer's labeling). Prolonged with continuous infusions >24 hours, heart failure, liver disease, shock, severe renal impairment, neonatal therapeutic hypothermia (van den Broek 2013; manufacturer's labeling).

Excretion: Urine (<10% as unchanged drug, ~90% as metabolites).

Patient Counseling Points

(For additional information see "Lidocaine (local and regional anesthetic) and (systemic): Patient drug information")

What is this drug used for?

- It is used to numb an area before a procedure.
- It is used to treat certain types of abnormal heartbeats.
- It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

Zingo:

- Pinpoint red spots on skin

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

All products:

- Methemoglobinemia like a blue or gray color of the lips, nails, or skin; a heartbeat that does not feel normal; seizures; severe dizziness or passing out; severe headache; feeling very sleepy; feeling tired or weak; or shortness of breath
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Zingo:

- Burning
- Bruising
- Bleeding where the shot is given

All other products:

- Too much acid in the blood (acidosis) like confusion; fast breathing; fast heartbeat; a heartbeat that does not feel normal; very bad stomach pain, upset stomach, or throwing up; feeling very sleepy; shortness of breath; or feeling very tired or weak
- Trouble breathing, slow breathing, or shallow breathing
- Feeling lightheaded, sleepy, confused, or having blurred eyesight
- Slow heartbeat
- Feeling nervous and excitable
- Trouble speaking
- A burning, numbness, or tingling feeling that is not normal
- Restlessness
- Anxiety
- Change in eyesight
- Ringing in ears
- Dizziness or passing out
- Very bad headache
- Twitching
- Shakiness
- Seizures
- Feeling hot or cold
- Metallic taste
- Chest pain
- Mood changes

Injection (if given into the spine):

- Not able to get or keep an erection
- Not able to move
- Not able to control stools or urine
- Trouble passing urine

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Xylocaine | Xylocard;
- (AT) Austria: Xylocard;
- (AU) Australia: Lignocaine | Xylocaine | Xylocard;
- (BD) Bangladesh: Lidocaine;
- (BE) Belgium: Lidocaine accord;
- (BG) Bulgaria: Lidocain;
- (BR) Brazil: Xylocaina;
- (CN) China: Li shu ka | Lidocaine | Lidocaine aerosol (ii);
- (CO) Colombia: Farmacaina | Lidocaina | Roxicaina | Xylocaina;
- (DE) Germany: Lidocain Braun | Lidocain hcl B. Braun | Lidocard | Min-I-Jet Lidocain | Ryligency | Xylocain | Xylonor n;
- (EC) Ecuador: Roxicaina;
- (EE) Estonia: Xylocaine;
- (EG) Egypt: Lidocaine;
- (ES) Spain: Xilonibsa;
- (FI) Finland: Lidocard | Xylocain | Xylocard;
- (FR) France: Ryligency | Xylocard;
- (GB) United Kingdom: Lignocaine | Sure | Xylocard;
- (HK) Hong Kong: Xylocaine | Xylocard;
- (HU) Hungary: Lidocain;
- (IE) Ireland: Xylocard;
- (IL) Israel: Stud 100;
- (IQ) Iraq: Lidocaine p;
- (IT) Italy: Lidocaina abbott;
- (JO) Jordan: Avocaine;
- (JP) Japan: Lidocaine takata | Olives | Olives k | Olives takata | Xylocaine;
- (KR) Korea, Republic of: Xylocaine;
- (KW) Kuwait: Lidocaine HCL | Ryligency | Xylocard | Xylonor;
- (LB) Lebanon: Lidocaine | Procomil | Xylocaine;
- (LV) Latvia: Xylocard;
- (MX) Mexico: Mortelec;
- (MY) Malaysia: Xylocard;
- (NL) Netherlands: Xylocaine | Xylocard;
- (NO) Norway: Lidokain sad | Lignocaine hydrochloride fresenius | Xylocard;
- (NZ) New Zealand: Xylocaine plain | Xylocard;
- (PE) Peru: Lidocaina | Roxicaina | Xilonest s.p. | Xylocaina;
- (PH) Philippines: Xylocaine;
- (PK) Pakistan: Lignocaine;
- (PL) Poland: Lidocain | Lignocainum | Xylocain;
- (PR) Puerto Rico: Aspercreme with lidocaine | Lidocaine HCL | Xylocaine cardiac;
- (PT) Portugal: Lidocaina | Lidocaina braun | Lidoject | Xylocard;
- (PY) Paraguay: Lidocaina veinfar;
- (QA) Qatar: Lignoc | Medocaine | Zylodine;
- (RU) Russian Federation: Lidocain;
- (SA) Saudi Arabia: Xylocaine | Xylocard;
- (SE) Sweden: Xylocain | Xylocard;
- (SG) Singapore: Xylocard;
- (SI) Slovenia: Xylocain;
- (SR) Suriname: Lidocaine hcl cf;
- (TH) Thailand: Xylocaine | Xylocard;
- (TW) Taiwan: Lidocain | Lidocaine | Lignocaine | Xylocaine;
- (UA) Ukraine: Lidocain;
- (UY) Uruguay: Xylo Efa Cardiology;

- (VE) Venezuela, Bolivarian Republic of: Farmacaina

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